DEVOPS

LAB 05

Build, Test, and Package Applications with IntelliJ IDEA and Docker

Name: Fatima Noor Ul Imran

Registration Number: 2023-BSCS-33

Course Code: BCS-649

Section: A

GitHub Username: fatimanoorulimran

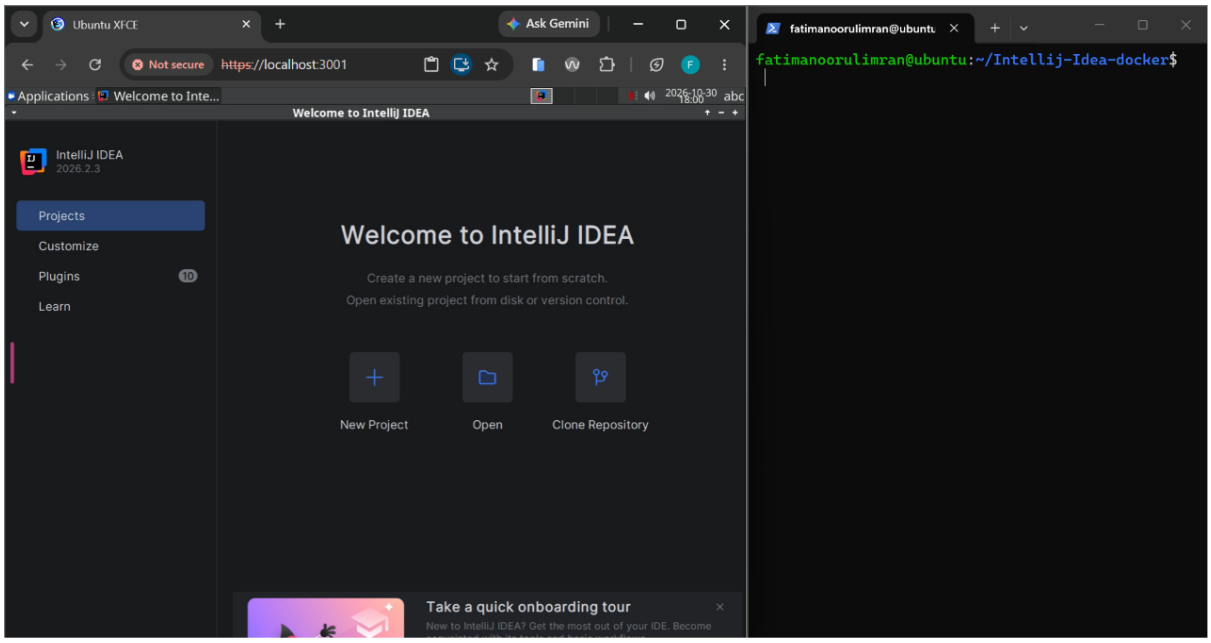
Task 1: Clone and Launch IntelliJ IDEA

```
fatimanoorulimran@ubuntu:~$ cd ~
fatimanoorulimran@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Intellij-Idea-docker.git
Cloning into 'Intellij-Idea-docker'...
remote: Enumerating objects: 66, done.
remote: Counting objects: 100% (66/66), done.
remote: Compressing objects: 100% (59/59), done.
remote: Total 66 (delta 29), reused 5 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (66/66), 26.98 KiB | 2.45 MiB/s, done.
Resolving deltas: 100% (29/29), done.
fatimanoorulimran@ubuntu:~$ cd Intellij-Idea-docker
fatimanoorulimran@ubuntu:~/Intellij-Idea-docker$
```

Cloned the IntelliJ Docker repository on the Ubuntu host.

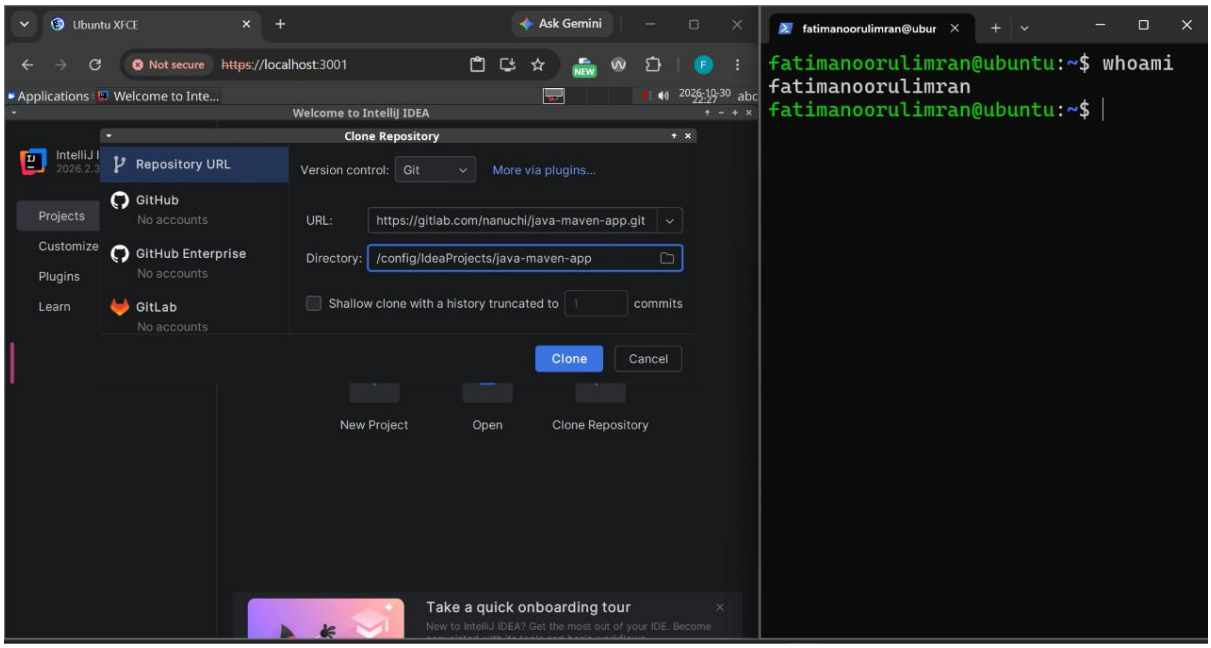
```
fatimanoorulimran@ubuntu:~/Intellij-Idea-docker$ docker compose ps
NAME                              IMAGE                          COMMAND   SERVICE    CREATED       STATUS              POR
TS
intellij-idea-docker-intellij-1   intellij-idea-docker-intellij   "/init"   intellij   5 hours ago   Up About a minute   300
0/tcp, 127.0.0.1:3001->3001/tcp
fatimanoorulimran@ubuntu:~/Intellij-Idea-docker$
```

docker compose ps shows the intellij service running with desktop port 3001.

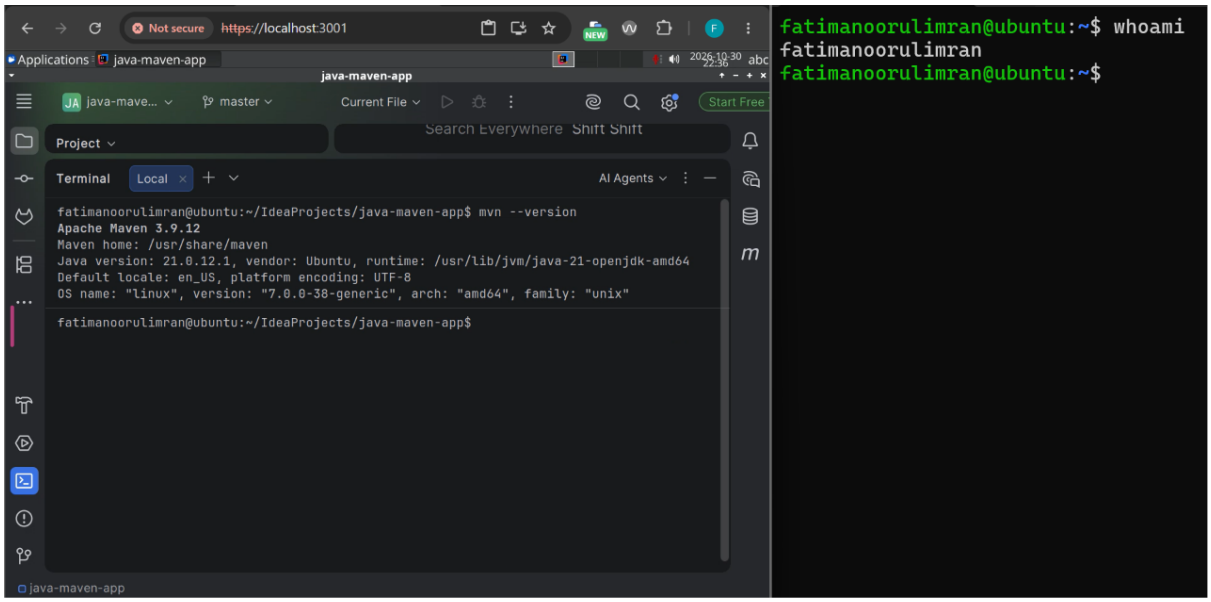


IntelliJ IDEA Welcome screen in the browser desktop, with the Ubuntu host terminal beside it.

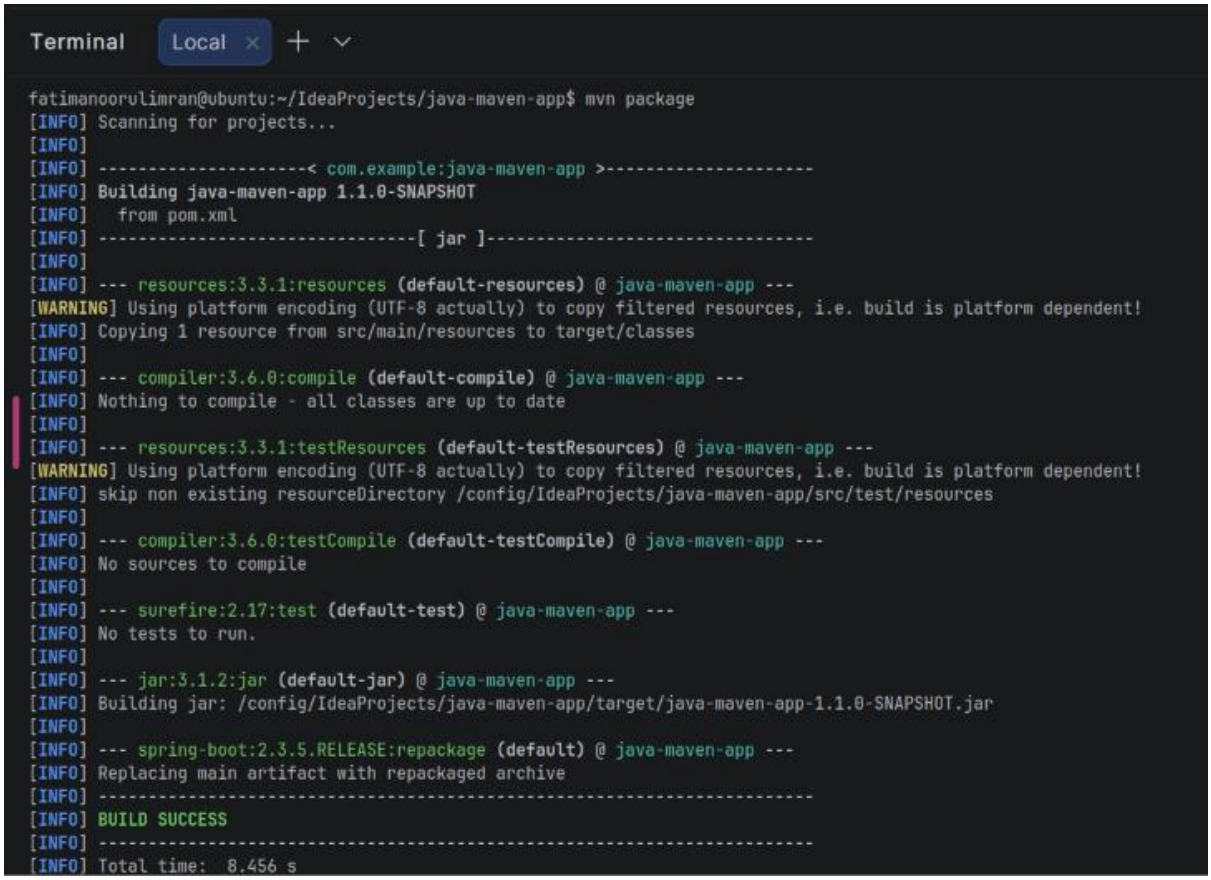
Task 2: Build and Run the Maven Project



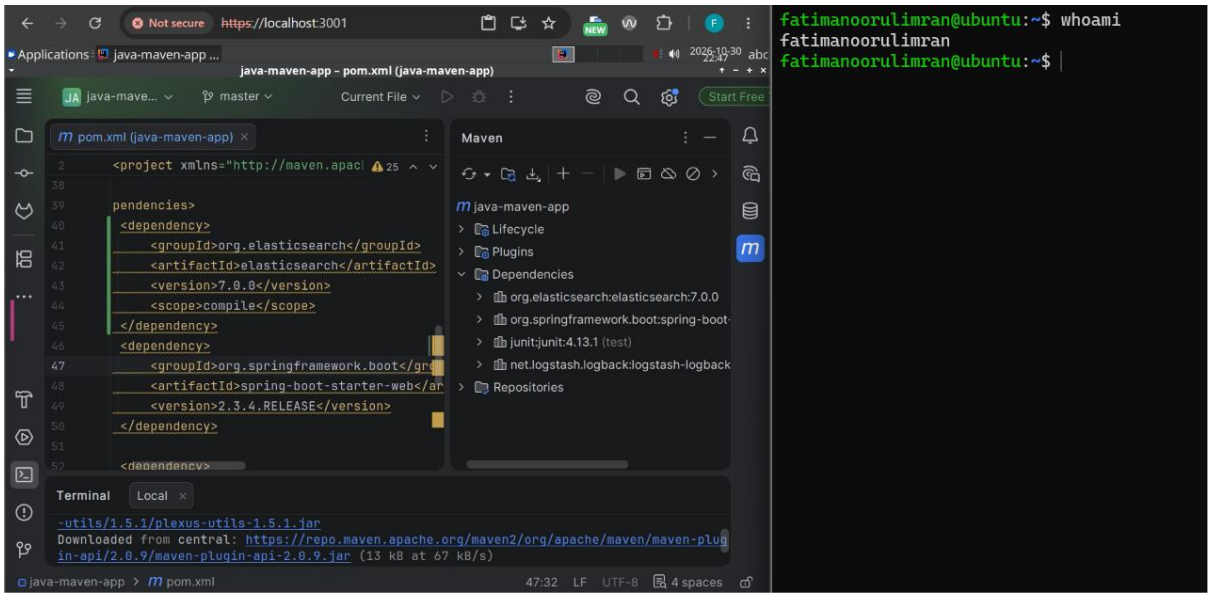
Clone dialog with the java-maven-app URL and destination directory.



Maven installed and verified with mvn --version in IntelliJ's terminal.



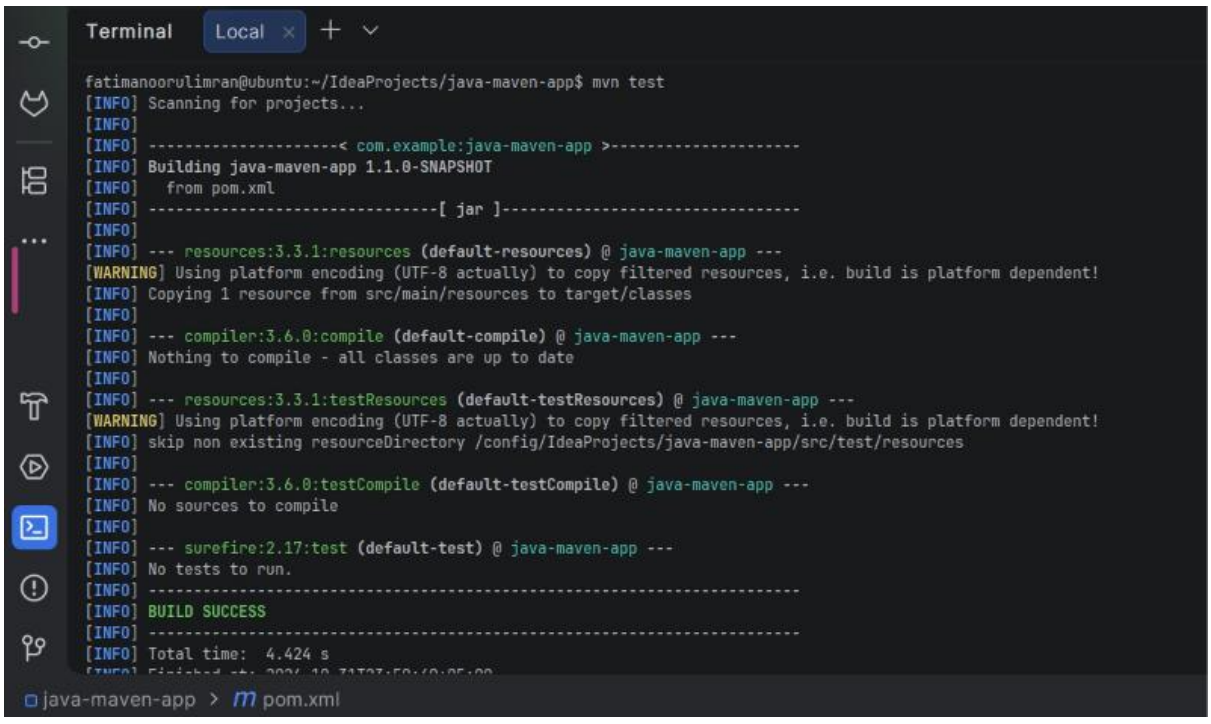
mvn package completed with BUILD SUCCESS.



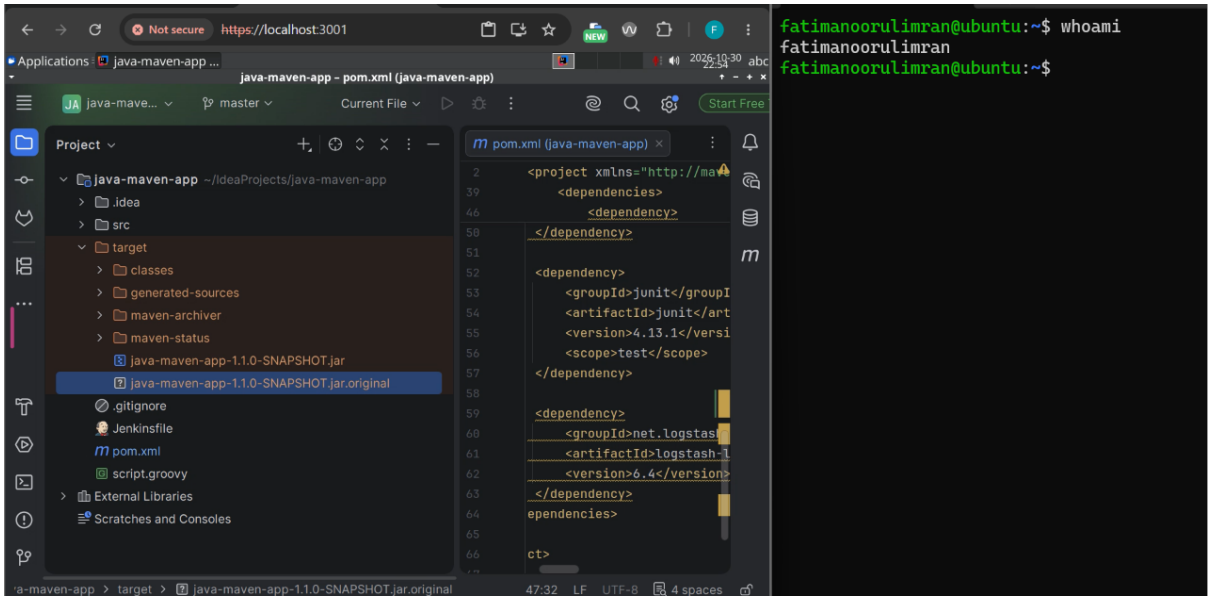
Elasticsearch 7.0.0 dependency added to pom.xml and synchronized in the Maven tool window.

```
fatimanoorulimran@ubuntu:~/IdeaProjects/java-maven-app$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] --------------------< com.example:java-maven-app >--------------------
[INFO] Building java-maven-app 1.1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.6.0:compile (default-compile) @ java-maven-app ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ java-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /config/IdeaProjects/java-maven-app/src/test/resources
[INFO]
[INFO] --- compiler:3.6.0:testCompile (default-testCompile) @ java-maven-app ---
[INFO] No sources to compile
[INFO]
[INFO] --- surefire:2.17:test (default-test) @ java-maven-app ---
[INFO] No tests to run.
[INFO]
[INFO] --- jar:3.1.2:jar (default-jar) @ java-maven-app ---
[INFO] Building jar: /config/IdeaProjects/java-maven-app/target/java-maven-app-1.1.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:2.3.5.RELEASE:repackage (default) @ java-maven-app ---
[INFO] Replacing main artifact with repackaged archive
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  6.040 s
```

mvn package completed with BUILD SUCCESS after adding the dependency.

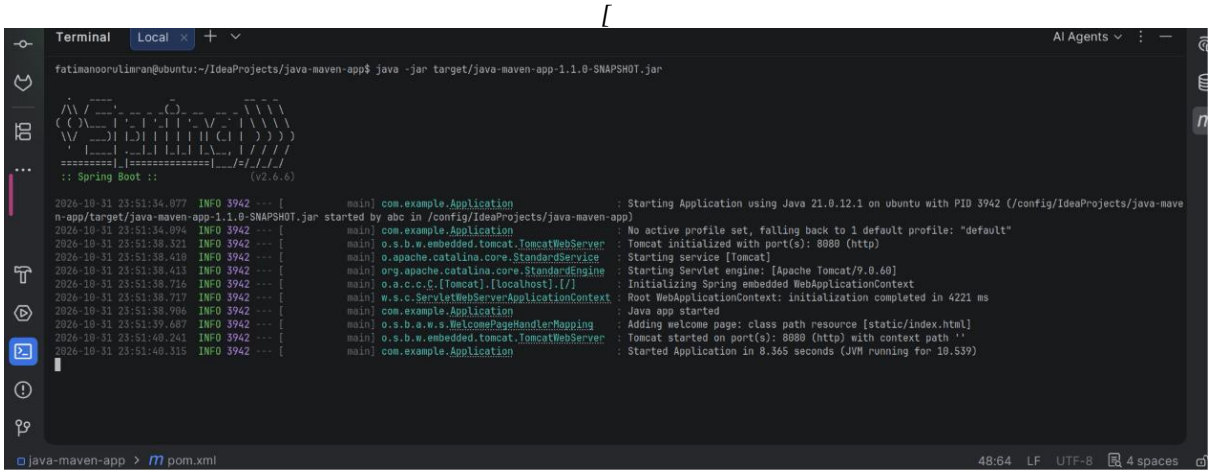


mvn test ran the test phase; no tests were found and the build succeeded.

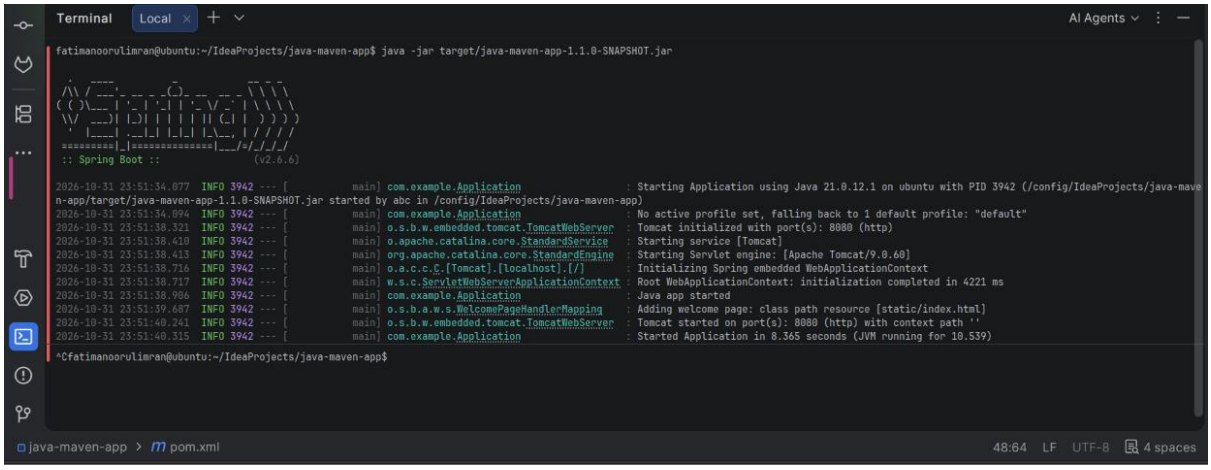


The generated java-maven-app-1.1.0-SNAPSHOT.jar in the target directory.

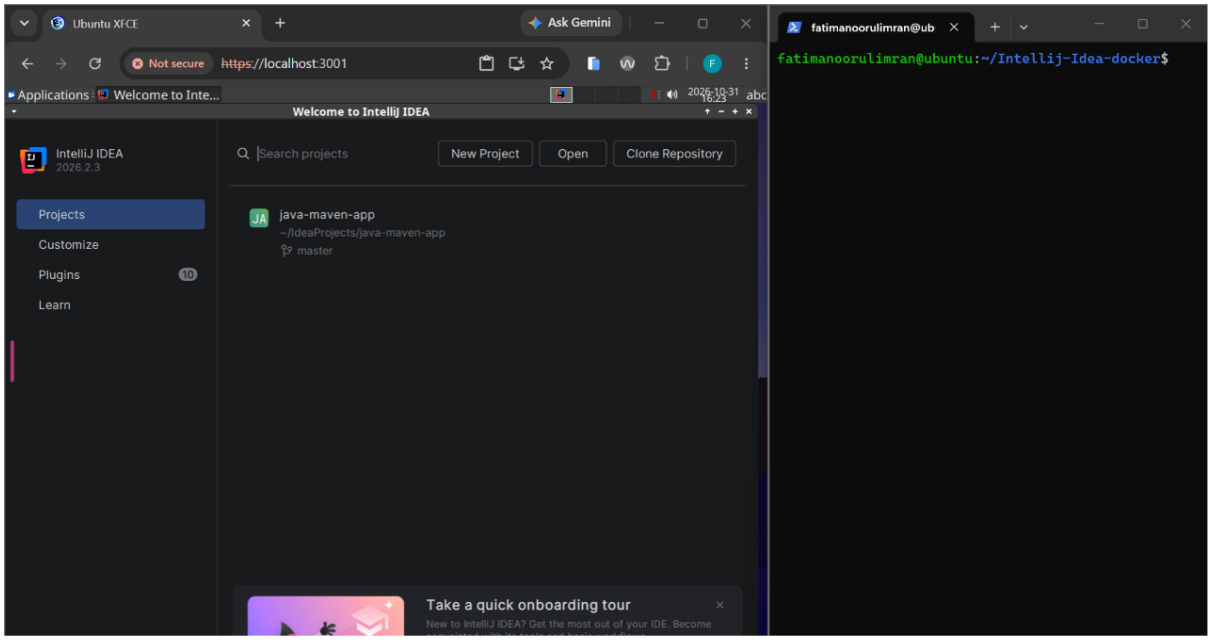
[



The JAR started with java -jar and Tomcat started on port 8080.

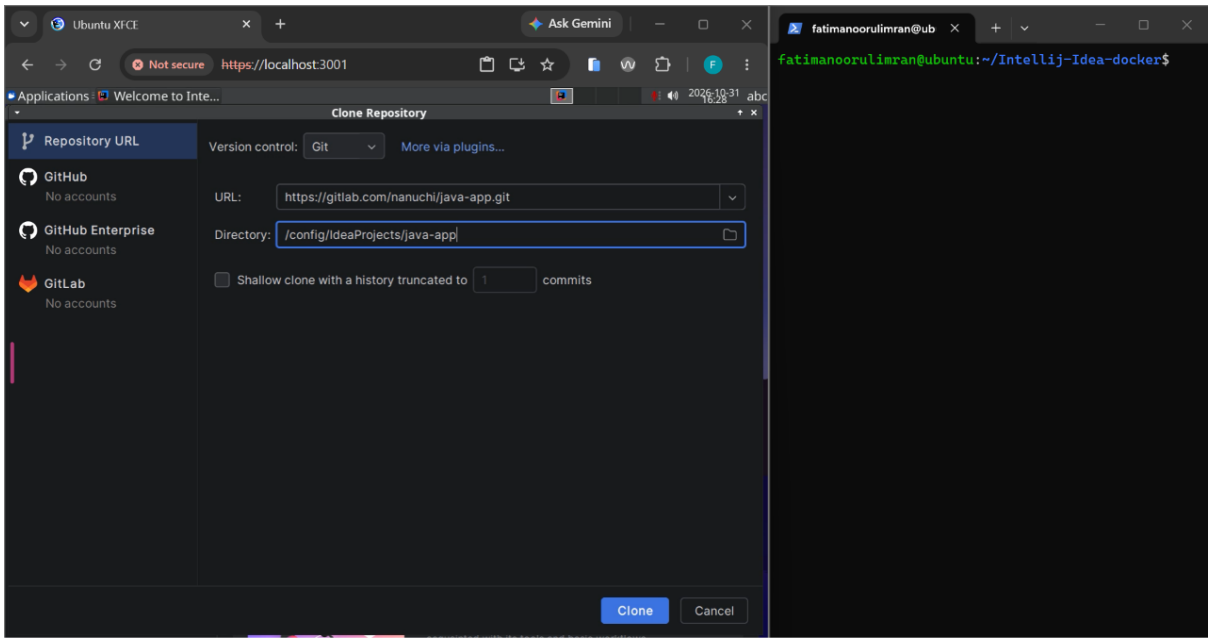


The application stopped with Ctrl+C and the prompt returned.

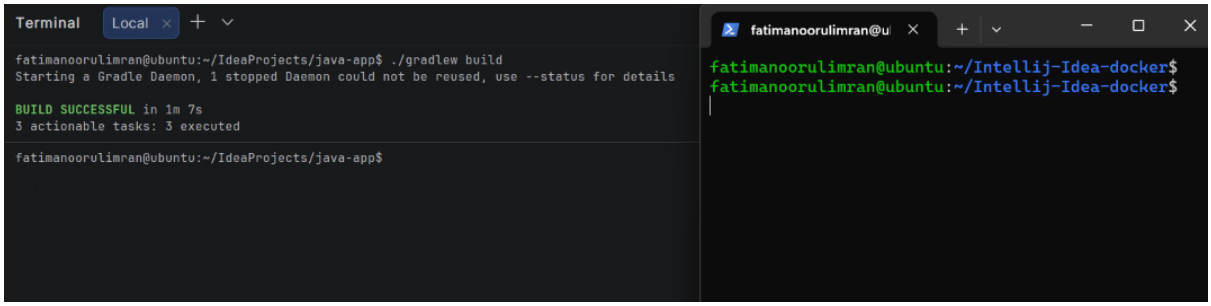


Project closed and IntelliJ Welcome screen shown beside the host terminal.

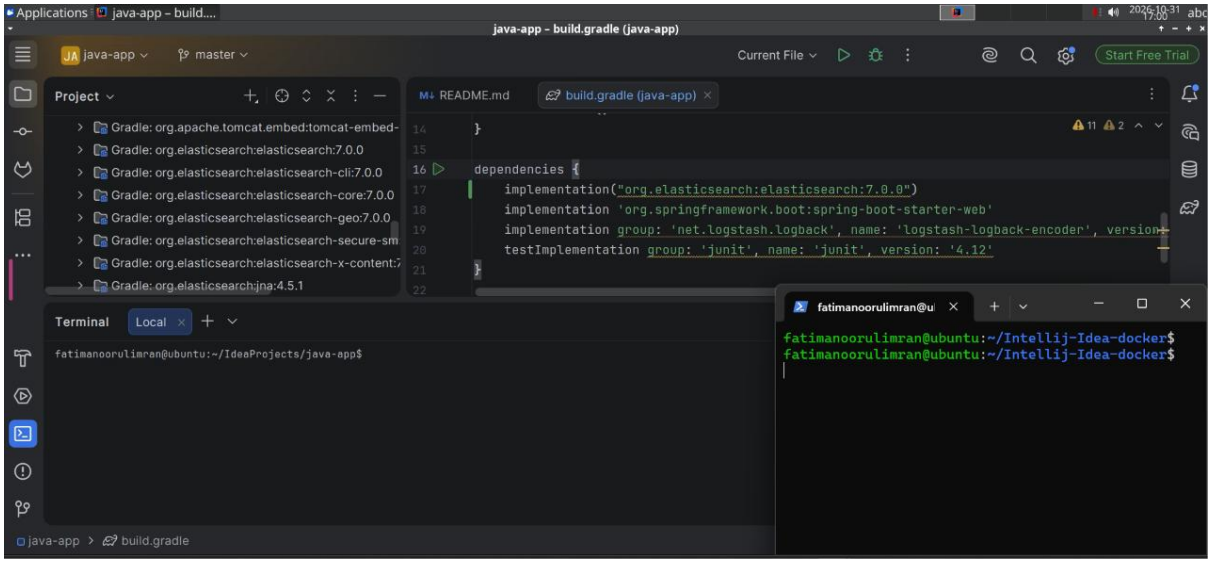
Task 3: Build and Run the Gradle Project



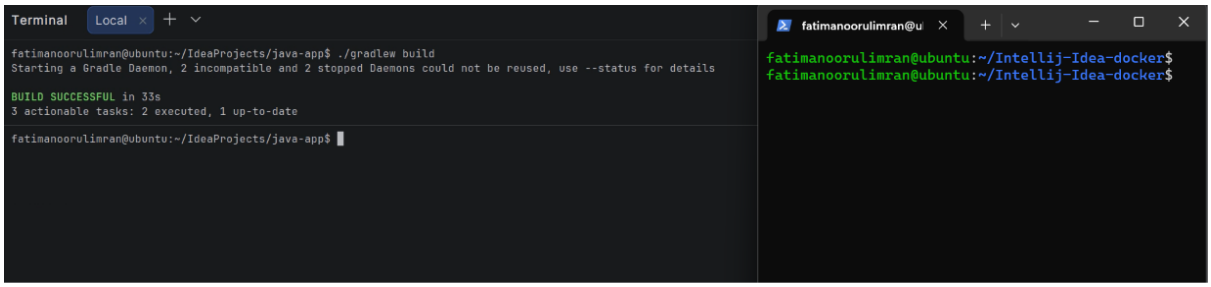
Clone dialog with the java-app URL and destination directory.



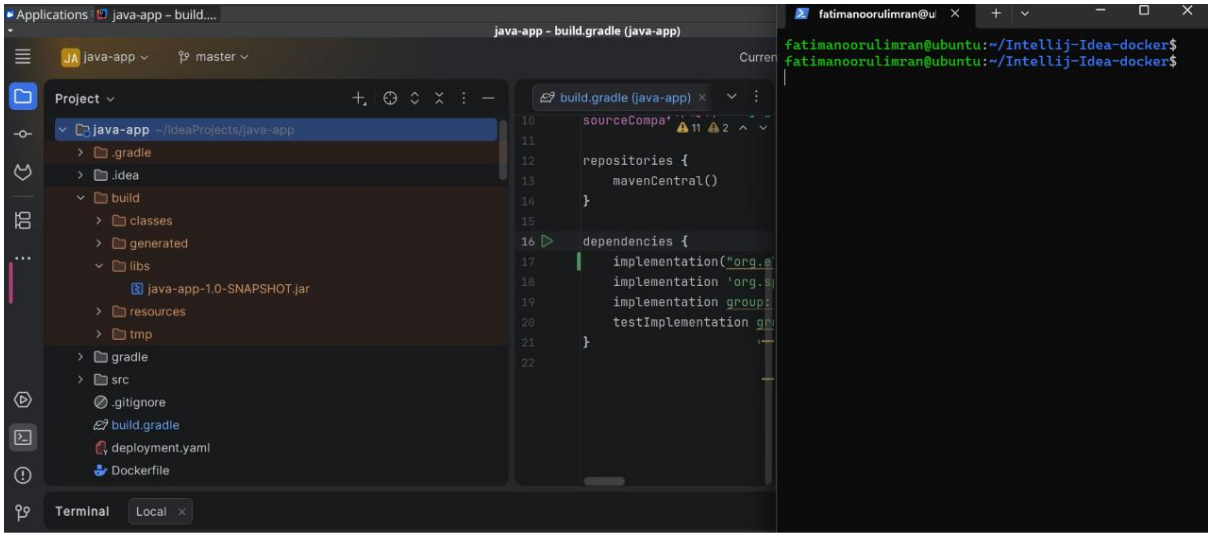
./gradlew build completed with BUILD SUCCESSFUL.



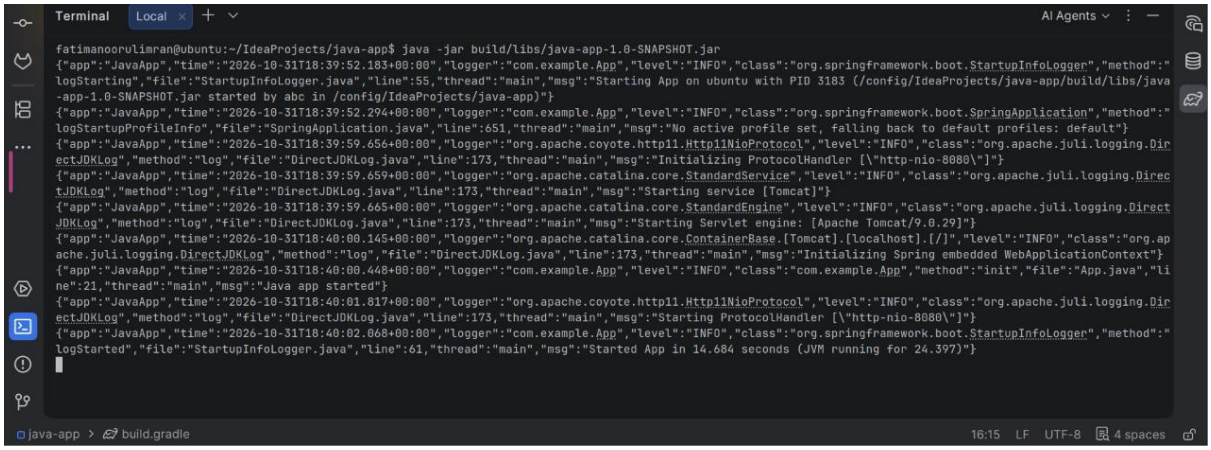
Elasticsearch 7.0.0 added to build.gradle and visible under External Libraries after the Gradle sync.



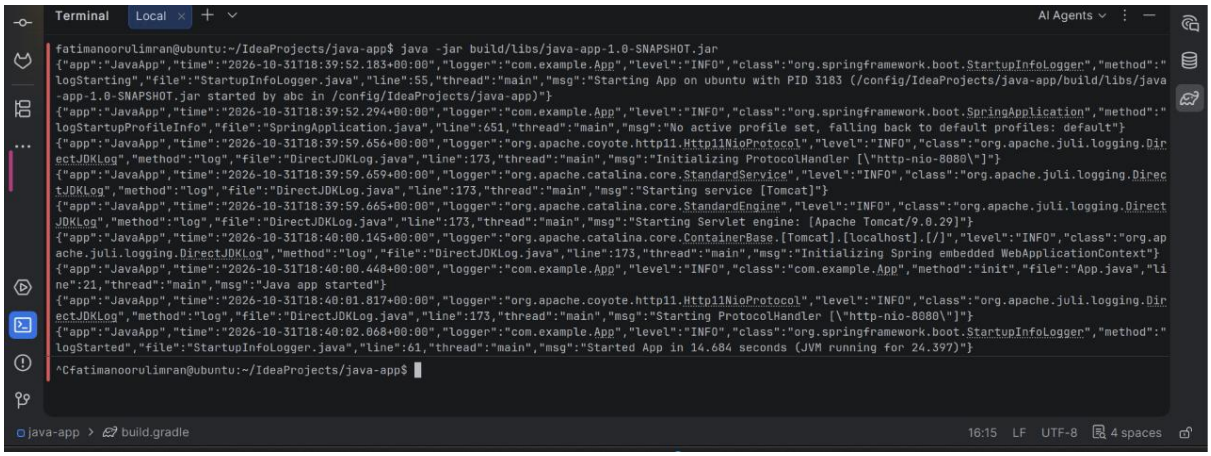
./gradlew build completed with BUILD SUCCESSFUL after adding the dependency.



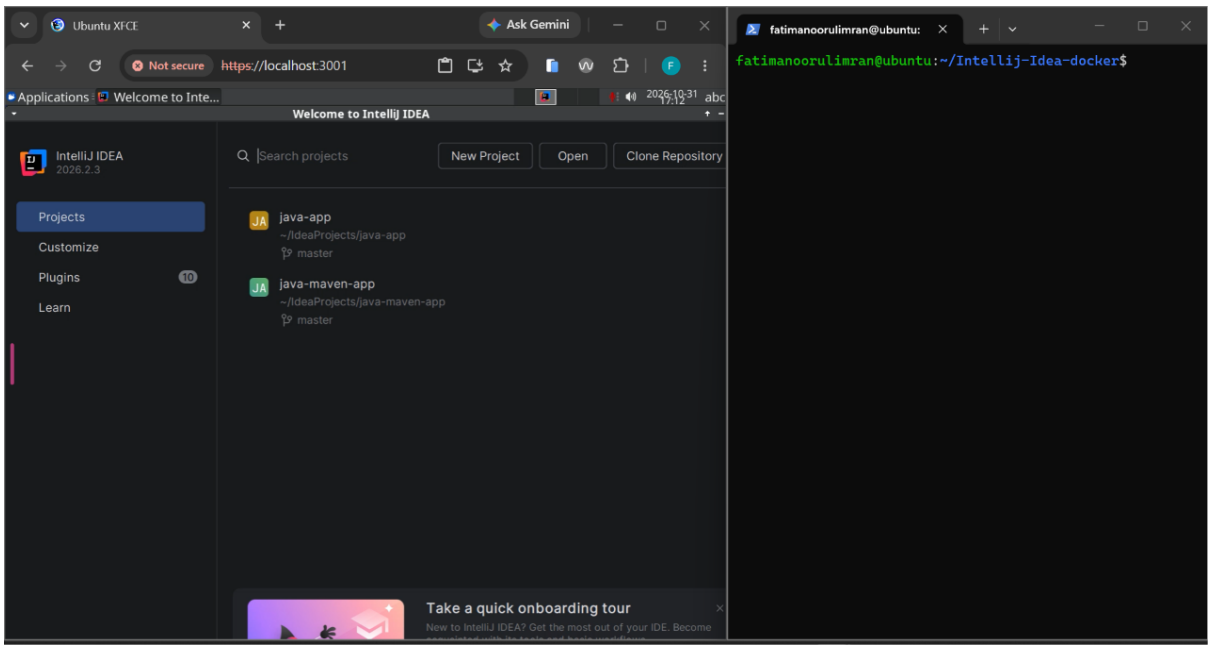
The generated java-app-1.0-SNAPSHOT.jar in build/libs.



The JAR started with java -jar and the application started successfully.

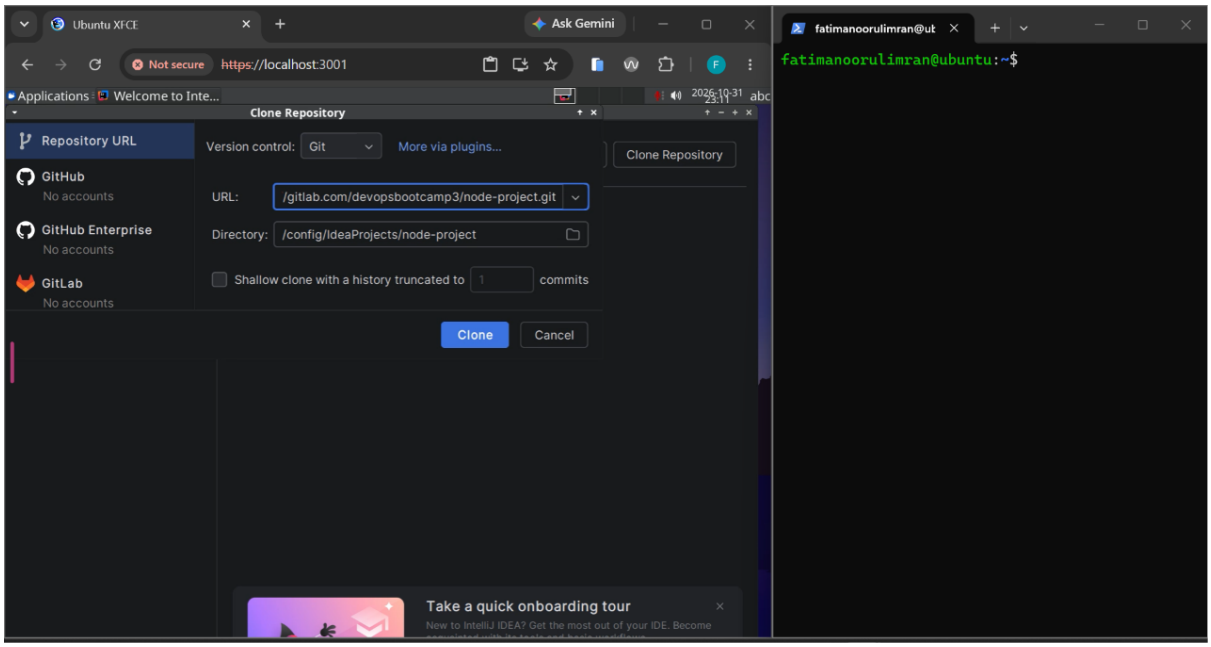


The application stopped with Ctrl+C and the prompt returned.

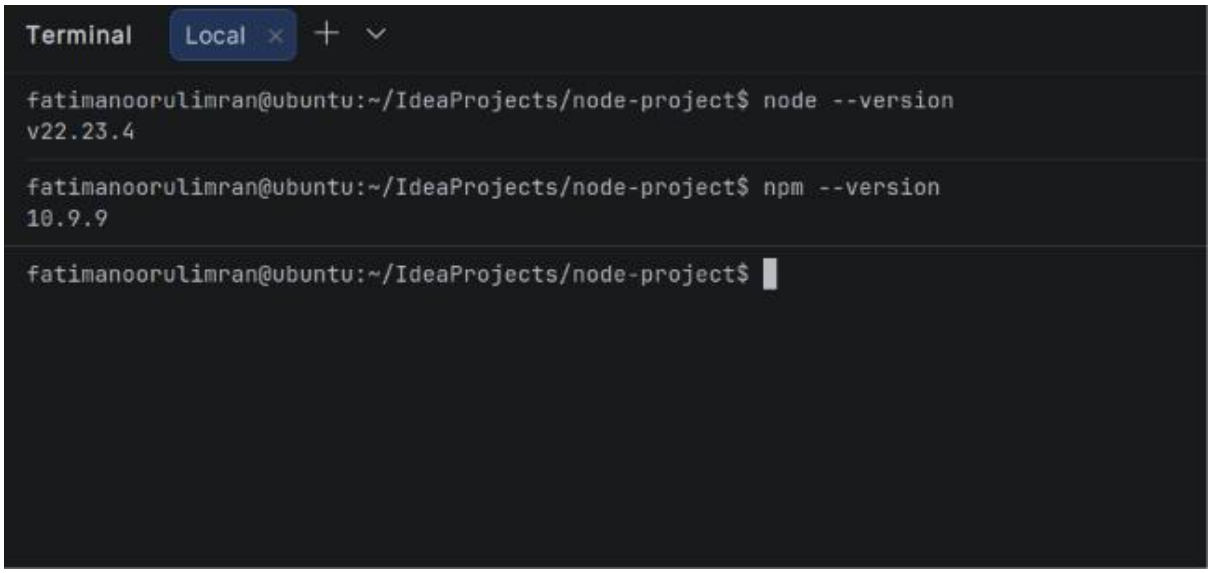


Project closed and IntelliJ Welcome screen shown beside the host terminal.

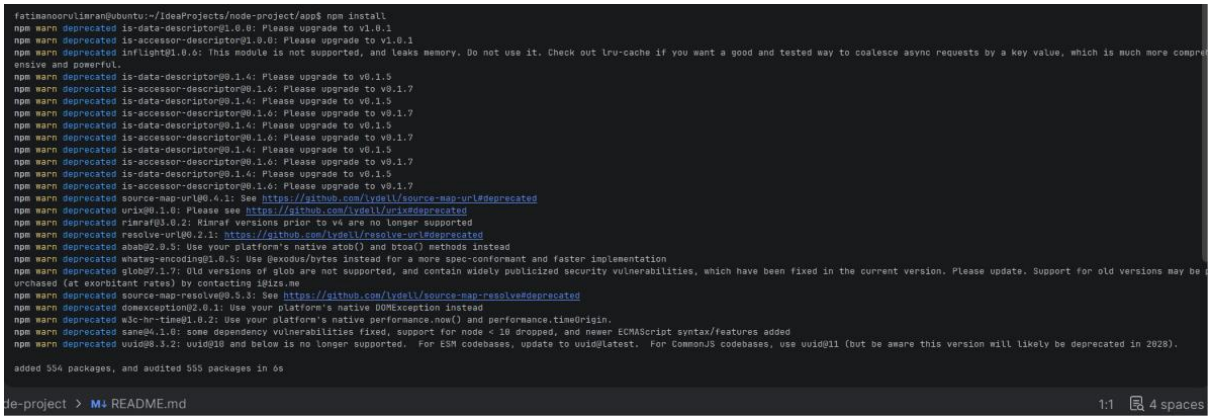
Task 4: Install and Run the Node.js Project



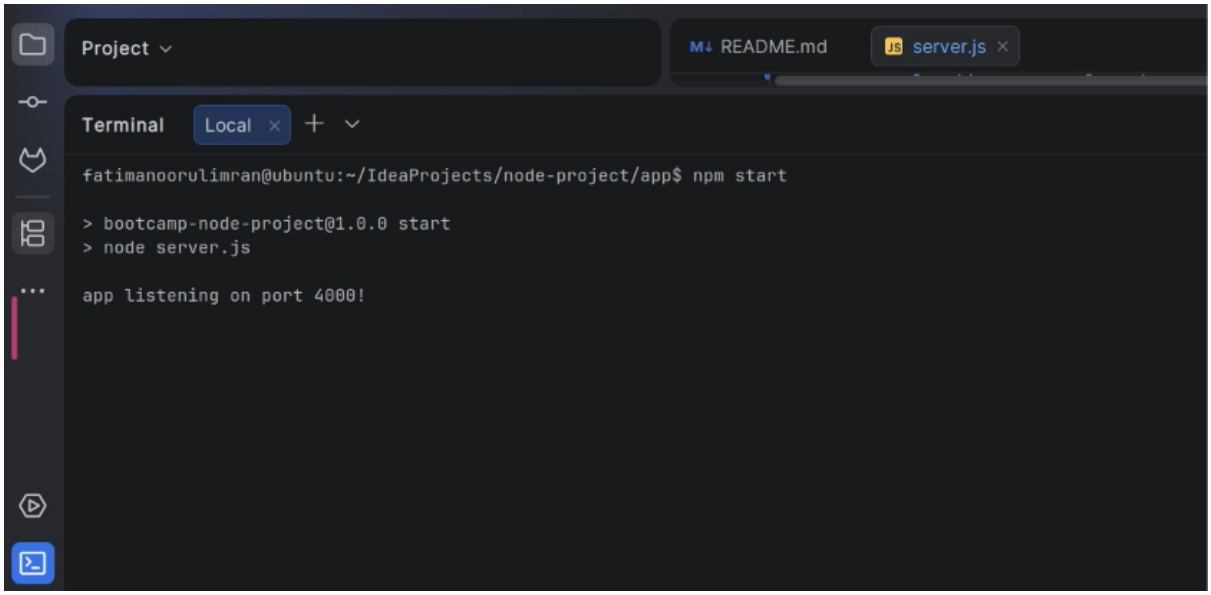
Clone dialog with the node-project URL and destination directory.



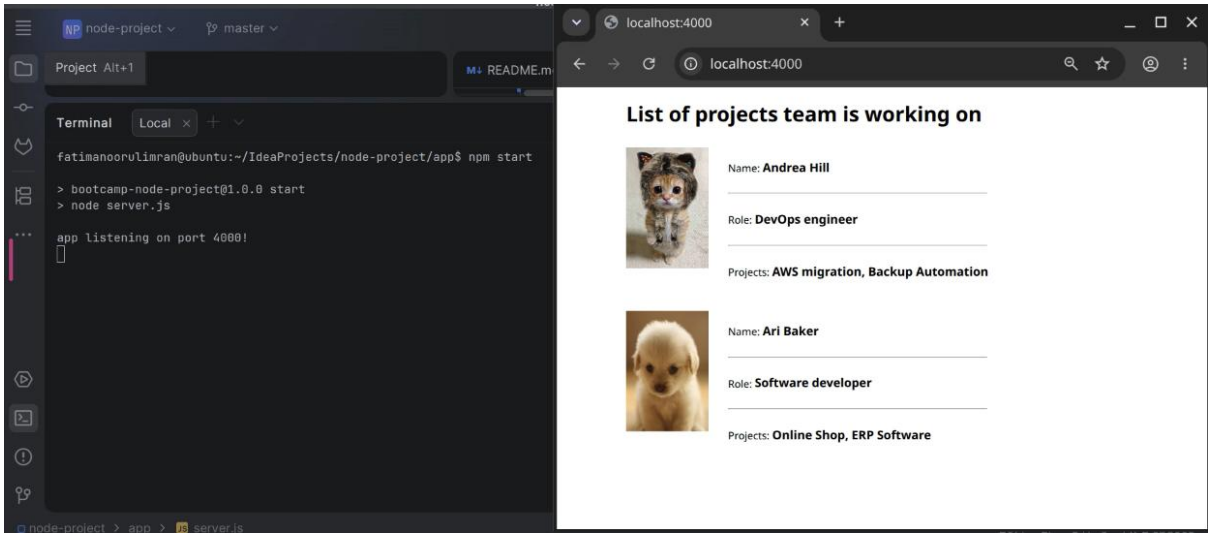
Node.js and npm installed, with both versions shown.



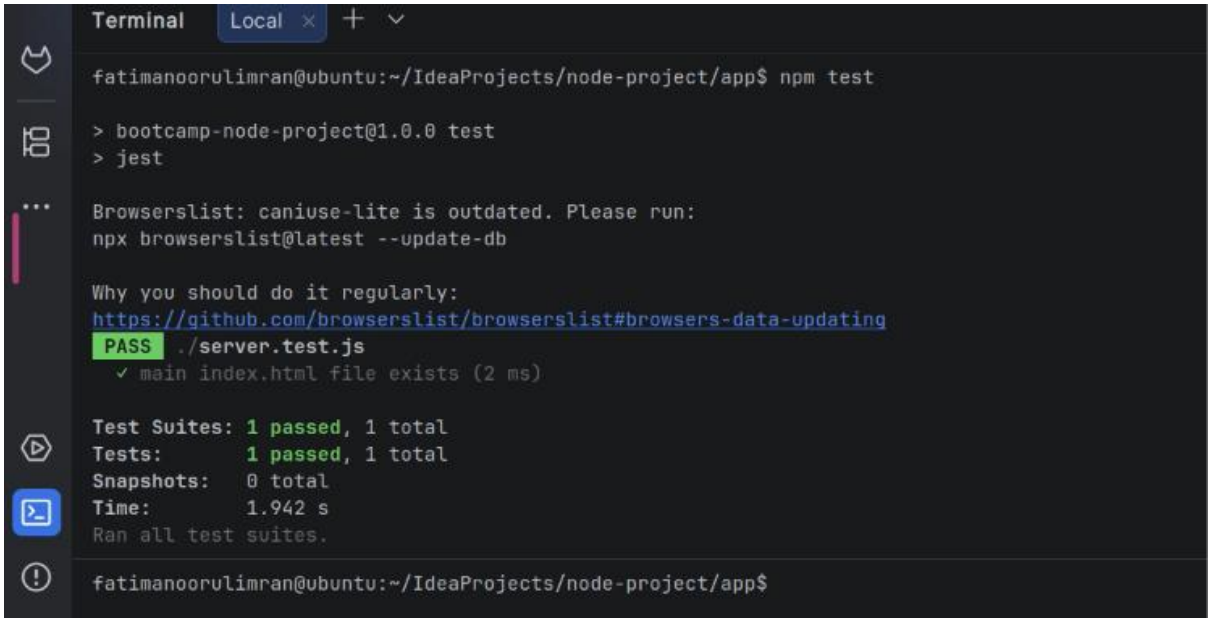
npm install completed in the app directory.



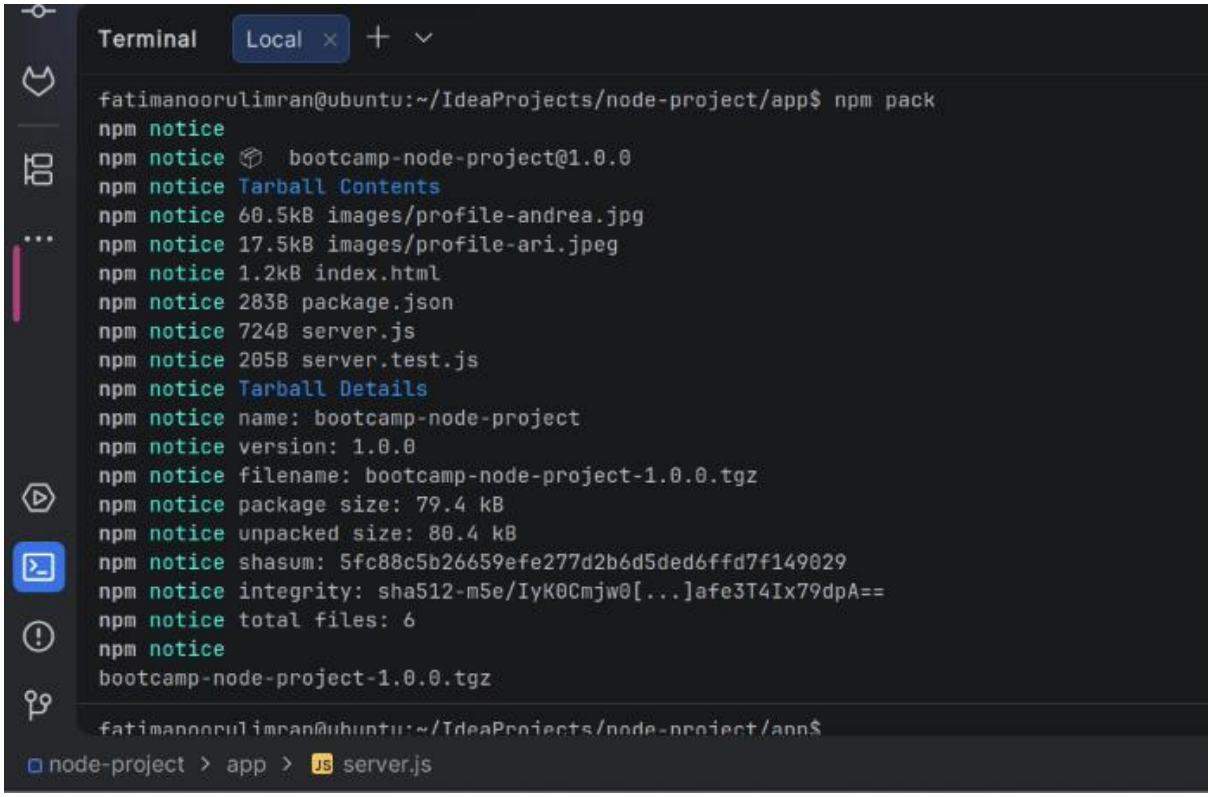
npm start running after changing the port to 4000: app listening on port 4000!



The Node.js project page open at http://localhost:4000 in the container desktop browser.

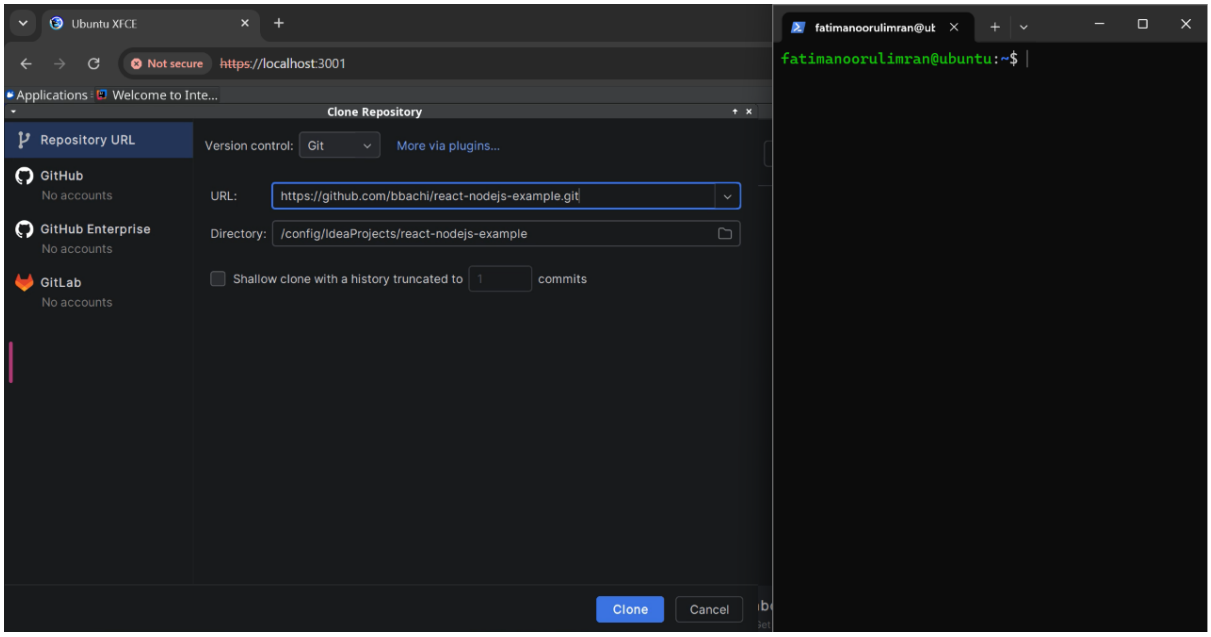


npm test ran the Jest test and it passed.

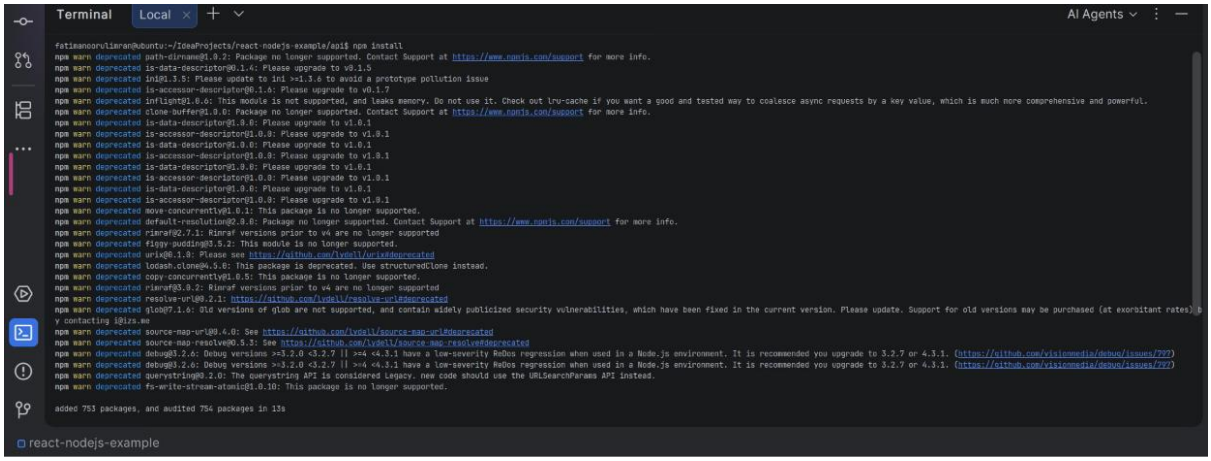


npm pack created bootcamp-node-project-1.0.0.tgz.

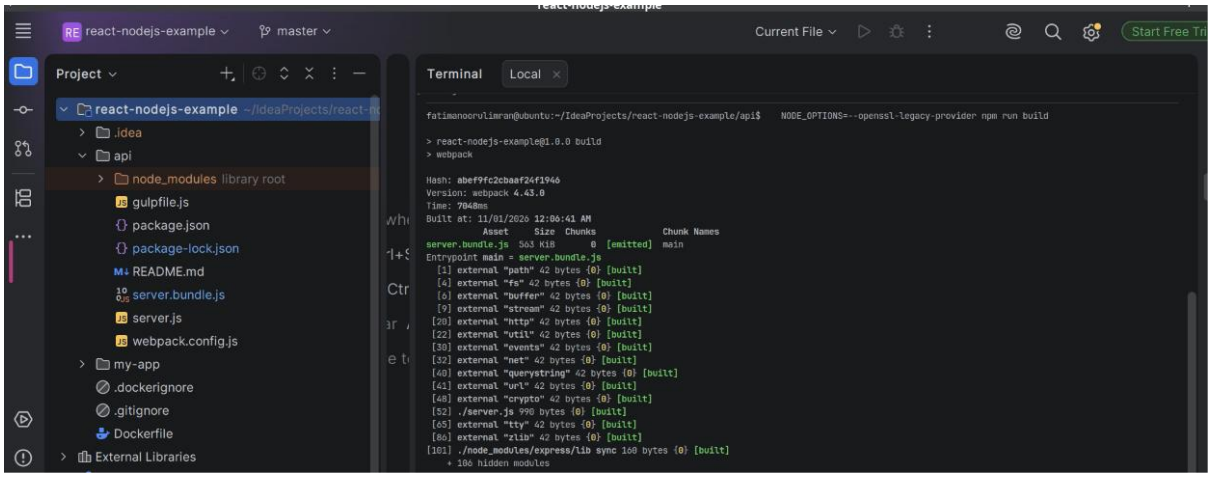
Task 5: Build the React and Node.js API Artifact



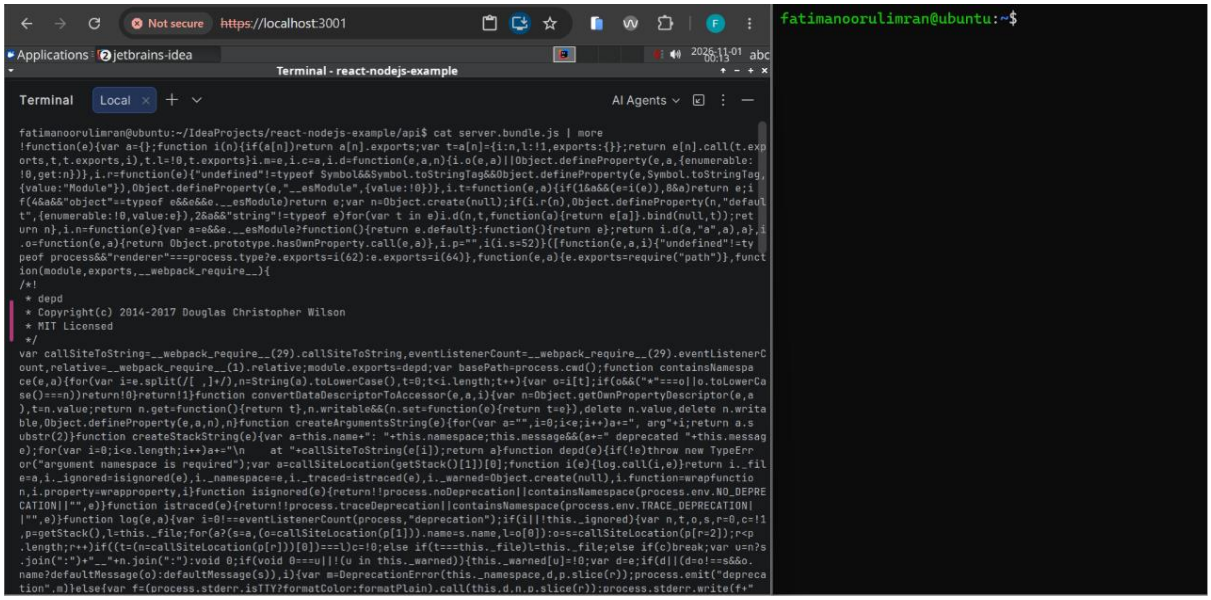
Clone dialog with the react-nodejs-example URL and destination directory.



npm install completed in the api directory and node_modules is present.

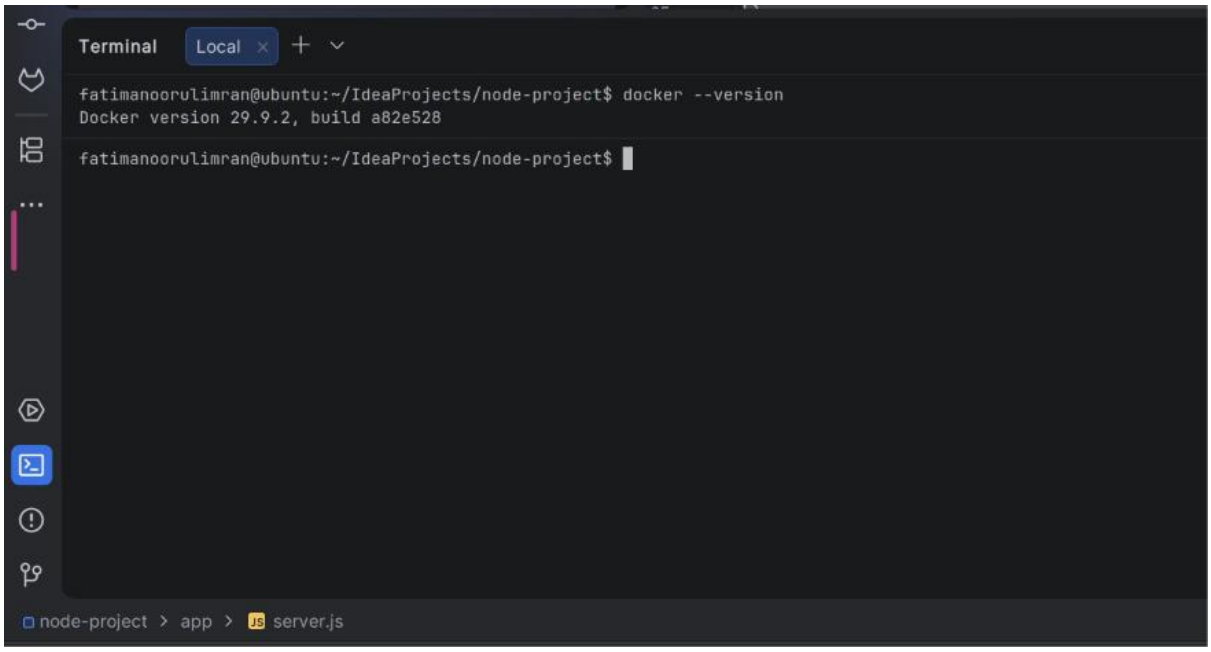


npm run build ran webpack and generated server.bundle.js in the api directory.

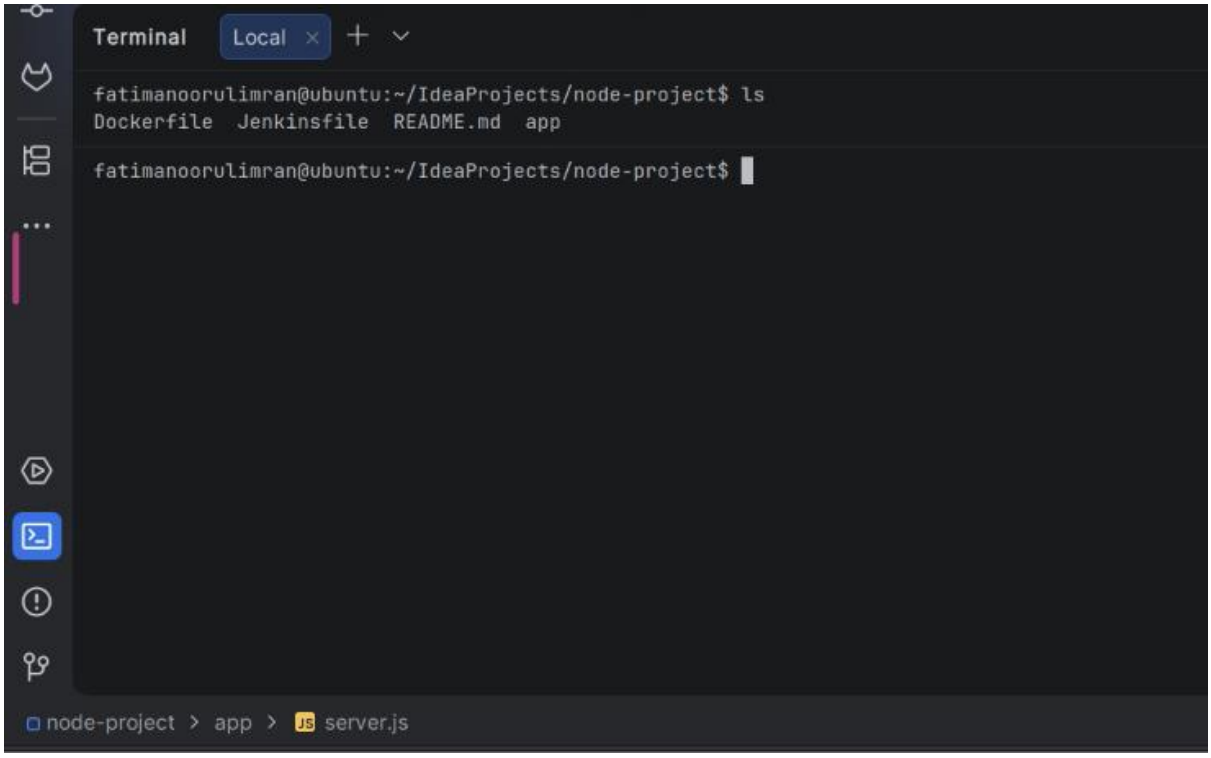


The bundled server.bundle.js viewed with cat server.bundle.js | more.

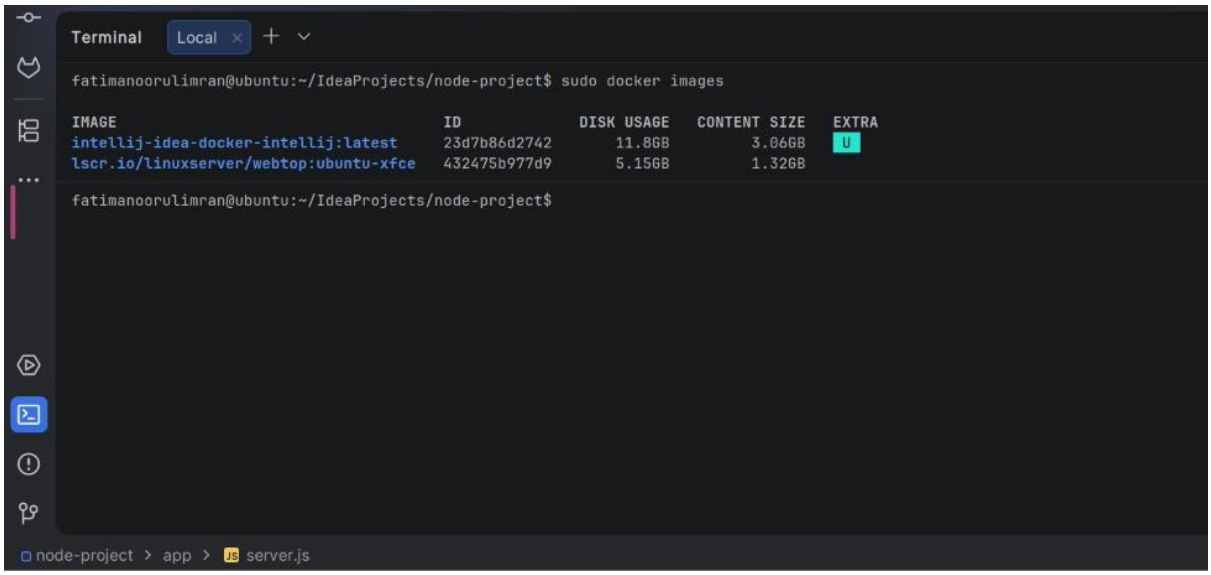
Task 6: Build a Docker Image for the Node.js Project



Docker installed, with its version shown.



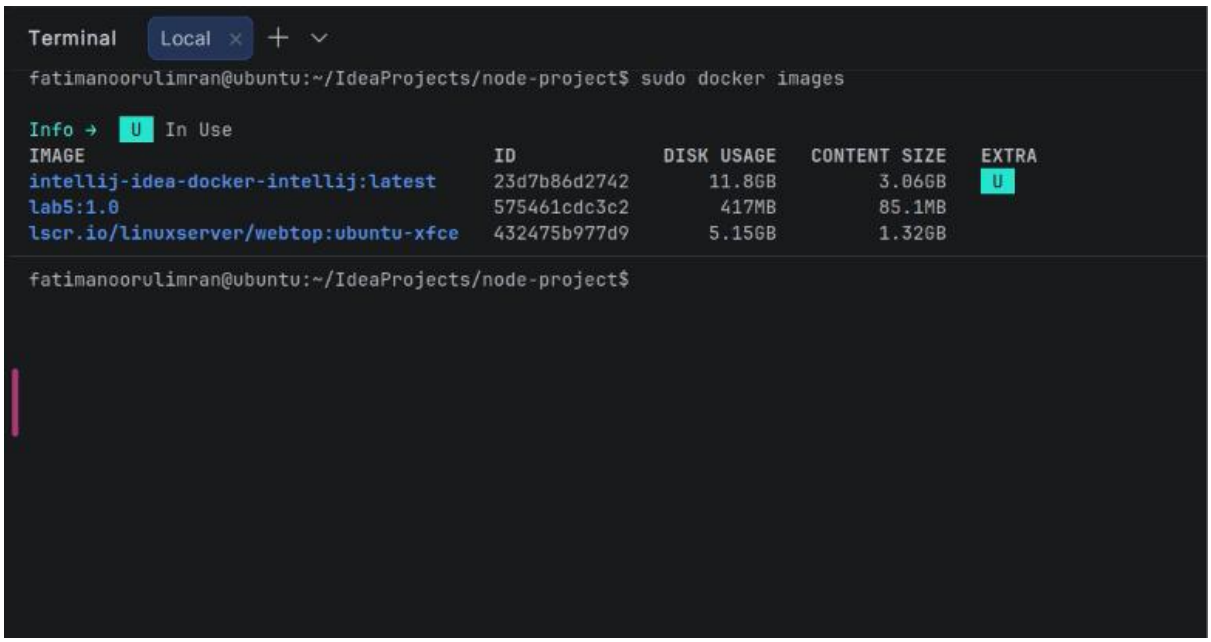
ls shows the Dockerfile in the node-project root directory.



Docker images listed before the build.



docker build -t lab5:1.0 . completed successfully.



Docker images listed after the build, showing repository lab5 with tag 1.0.